

Seminar Notes on Kodaira's Complex Manifold Theory

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These notes supplement K. Kodaira, 複素多様体と複素構造の変形 *I* (1968), filling in derivations omitted in the original lecture notes.

Preface

These are my seminar notes from reading Kodaira's lecture notes 複素多様体と複素構造の変形 *I* (1968) during my undergraduate research. The original frequently states a result and leaves its verification to the reader; the purpose of these notes is to write out those omitted steps in full.

Conventions

- Manifolds are assumed second countable and Hausdorff unless stated otherwise.
- $\mathbb{C}, \mathbb{R}, \mathbb{Z}, \mathbb{N}$ denote the complex, real, integer, and natural numbers.
- $\bar{\partial}$ is the Dolbeault operator; $\mathcal{O}$ is the structure sheaf.
- $H^q(X, \mathcal{F})$ denotes sheaf cohomology unless noted otherwise.

Contents

Preface	ii
1 Holomorphic Functions	1
2 Complex Manifolds	2
3 Differential Forms, Vector Bundles, and Sheaves	3
4 Sheaf Cohomology and the Dolbeault Theorem	4
5 The Dolbeault Operator	5
5.1 The operator $\bar{\partial}$	5

Chapter 1

Holomorphic Functions

(Notes in preparation.)

Chapter 2

Complex Manifolds

(Notes in preparation.)

Chapter 3

Differential Forms, Vector Bundles, and Sheaves

(Notes in preparation.)

Chapter 4

Sheaf Cohomology and the Dolbeault Theorem

(Notes in preparation.)

Chapter 5

The Dolbeault Operator

This chapter illustrates the style of these notes: we take a statement the book leaves to the reader and write out the omitted verification in full.

5.1 The operator $\bar{\partial}$

Definition 5.1. Let $U \subseteq \mathbb{C}^n$ be open and $f \in C^\infty(U)$. The *Dolbeault operator* is

$$\bar{\partial}f = \sum_{j=1}^n \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j,$$

extended to (p, q) -forms by acting on coefficient functions and leaving the generators $dz_I \wedge d\bar{z}_J$ fixed.

Remark 5.2. Kodaira states $\bar{\partial}^2 = 0$ without proof. We supply it for functions; the general case follows by linearity and the Leibniz rule.

Proposition 5.3. For every $f \in C^\infty(U)$ we have $\bar{\partial}(\bar{\partial}f) = 0$.

Proof. By Definition 5.1,

$$\bar{\partial}f = \sum_j \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j.$$

Applying $\bar{\partial}$ again to this 1-form, which by Definition 5.1 means differentiating each coefficient,

$$\bar{\partial}(\bar{\partial}f) = \sum_k \sum_j \frac{\partial^2 f}{\partial \bar{z}_k \partial \bar{z}_j} d\bar{z}_k \wedge d\bar{z}_j.$$

The mixed second derivatives are symmetric in j, k (Schwarz's theorem, valid since f is smooth), while $d\bar{z}_k \wedge d\bar{z}_j = -d\bar{z}_j \wedge d\bar{z}_k$ is antisymmetric. Pairing the (j, k) and (k, j) terms,

$$\frac{\partial^2 f}{\partial \bar{z}_k \partial \bar{z}_j} d\bar{z}_k \wedge d\bar{z}_j + \frac{\partial^2 f}{\partial \bar{z}_j \partial \bar{z}_k} d\bar{z}_j \wedge d\bar{z}_k = 0,$$

so every term cancels and $\bar{\partial}(\bar{\partial}f) = 0$. □